

Tensor computations

Monotonicity of the average number of critical rank-one approximations

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Challenging because a dimension comparison of Gaussian critical-point expectations needs control beyond the known integral formula; specialist importance concerns a particular statistic of rank-one approximation landscapes.

Statement

For integers $p \geq 3$ and $2 \leq n_1 \leq \dots \leq n_p$, let $G \in \bigotimes_{i=1}^p \mathbb{R}^{n_i}$ have independent standard normal entries. Let X be the smooth manifold of nonzero real rank-one tensors in this space, and let $N(G)$ be the number of critical points on X of

$$Z \mapsto \|G - Z\|_F^2.$$

Each critical tensor is counted once, regardless of how its vector factors are scaled. Define $a(n_1, \dots, n_p) = \mathbb{E}[N(G)]$; the critical-point count is finite almost surely.

If

$$n_p - 1 > \sum_{i=1}^{p-1} (n_i - 1),$$

is it always true that

$$a(n_1, \dots, n_{p-1}, n_p) \leq a(n_1, \dots, n_{p-1}, n_p - 1)?$$

The expectation on the right uses a standard Gaussian tensor in the smaller space.

Relevance

Rank-one approximation algorithms navigate a landscape of stationary points. This conjecture predicts how the average number of such points changes when one mode becomes much larger than the others.

References

1. J. Draisma and E. Horobeț, *The average number of critical rank-one approximations to a tensor*, Linear Multilinear Algebra 64 (2016), 2498–2518. DOI; [primary preprint](#), Conjecture 1.3, p.3; §1 defines the Gaussian average and Theorem 1.1 gives an integral formula.
2. S. Friedland and G. Ottaviani, *The Number of Singular Vector Tuples and Uniqueness of Best Rank-One Approximation of Tensors*, Found. Comput. Math. 14 (2014), 1209–1242. DOI; [primary preprint](#), Theorem 1, for the corresponding complex critical-point count.

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Rechecked [Draisma–Horobeț, Conjecture 1.3](#), including the direction of the inequality, and searched by the authors, critical-rank-one counts and monotonicity. No proof or counterexample to the real Gaussian statement was located. The matrix case is excluded here, and stabilization of complex critical-point counts does not settle this expectation. No recent explicit primary reaffirmation was found.